

To address this, we propose *StreamMeCo*, an efficient training-free Long-Term *Stream Agent Memory Compression* framework, achieving the first memory compression of industrial-level streaming video agent and supporting efficient memory graph retrieval. Based on the connectivity of the memory graph, we design a dual-branch memory compression method. For isolated text nodes, we use the *Edge-free Minmax sampling (EMsampling)* module for compression. First, we cluster the text nodes into different groups, then for each cluster, we first select the node closest to the cluster center and add it to the selected set, then calculate the minimum distance between each unselected node and the selected set, choosing the farthest point. This process continues until the number of selected nodes meets the predefined requirement. For connected text nodes, we use the *Edge-aware Weighting pruning (EW-pruning)* module. By calculating the weight edge matrix between text nodes and face nodes or voice nodes, we determine the entity importance of each text node. We then combine the embedding similarity of the text nodes to select the most dissimilar text nodes, and ultimately determine the retained text nodes based on a comprehensive fusion score.

Additionally, to reduce the performance degradation caused by compressing the memory graph, inspired by the human memory mechanism, we design a *Time-decay Memory Retrieval (TMR)* mechanism for efficiently retrieving the memory graph. We calculate the total similarity between text nodes in the memory graph and the model’s query for each time segment. Based on the total similarity of each time segment, we dynamically allocate the number of text nodes to different segments. Furthermore, to simulate the memory decay mechanism of the human brain, we assign nonlinear time-varying decay to the text nodes of each time segment, thereby adaptively balancing long-term memory and recent critical information during the retrieval stage, and guaranteeing the retrieval of more memories from recent segments.

In summary, the main contributions of this paper are as follows:

- We propose *edge-free minmax sampling* module and *edge-aware weighting pruning* module, which efficiently compress the memory graph based on its structural properties.
- We introduce a novel *time-decay memory retrieval* mechanism, which has been experimen-

tally proven to significantly outperform previous retrieval frameworks.

- We demonstrate the effectiveness of *StreamMeCo* through extensive experiments. On three high-difficulty benchmark datasets, under approximately **70%** memory graph compression, the proposed method achieves a **1.87×** speedup in memory retrieval while delivering an average accuracy improvement of **1.0%**.

2 Related Work

2.1 Streaming Video Understanding

Streaming video understanding enables a model to continuously process incoming frames, and upon receiving a user question Q_t at time t , generate responses based on the accumulated content $C_{1:t}$. Existing methods fall into two main paradigms: visual token compression and KV cache compression. The former selects informative tokens either in pixel space or after feature encoding (Chen et al., 2025; Dorovatas et al., 2025; Zeng et al., 2025; Yao et al., 2025a; Li et al., 2025b; Huang et al., 2025b; Chen et al., 2024). While the latter maintains a short-term sliding window and a long-term memory stack at the LLM level to preserve information across temporal scales (Yang et al., 2025c; Kim et al., 2025; Ning et al., 2025; Di et al., 2025; Xu et al., 2025b). Recently, M3-Agent (Long et al., 2025) departs from these designs by integrating long-term agent memory into streaming video understanding, showing strong potential for modeling long-range temporal dependencies and maintaining entity consistency.

2.2 Long-Term Agent Memory

Long-term agent memory is designed to provide LLM-based agents with persistent memory beyond a single context window. Depending on the type of input it handles, it can be categorized into unimodal and multimodal forms. Unimodal memory (Yu et al., 2025a; Zhou et al., 2025; Chhikara et al., 2025a; Packer et al., 2023) typically maintains consistency across multi-turn reasoning by constructing structured representations, latent embeddings, or summaries. In contrast, multimodal memory (Yu et al., 2025b; Zhang et al., 2025; Yao et al., 2025b; Fan et al., 2025) integrates visual, linguistic, and other signals into a unified space, updating them to support stable long-term reasoning and decision-making. Building on this foundation, M3-Agent is the first to introduce long-